

Causality for Large Language Models



Max Planck Institute for
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ETH zürich



Meta



Can Large Language Models Infer Causation from Correlation?

ICLR 2024



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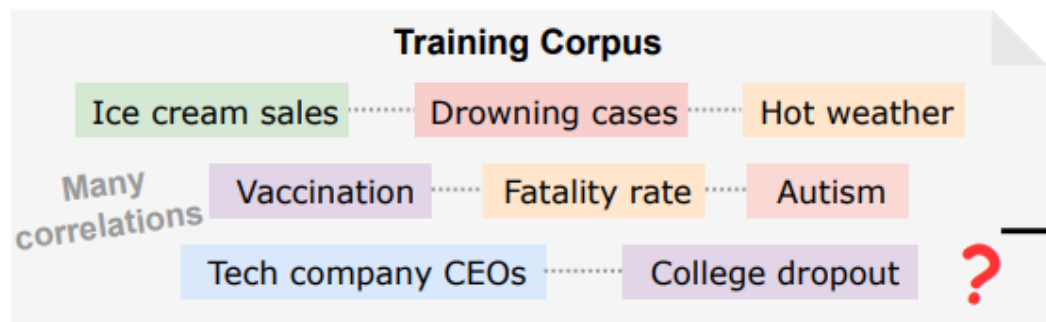
Bernhard
Schoelkopf†



207K samples

1K test data

1K dev data



?

How can LLMs process such information? **Correlation?**
Causation? What causes what?



We propose a new task: **Corr2Cause Inference**

This requires the skill of inferring causation from correlation

Suppose we know that A correlates with B.
Can we infer that A causes B?

No

A correlates with B. C correlates with B. However, A is independent of C.
Can we infer that A causes B?

Yes

Previous tasks:

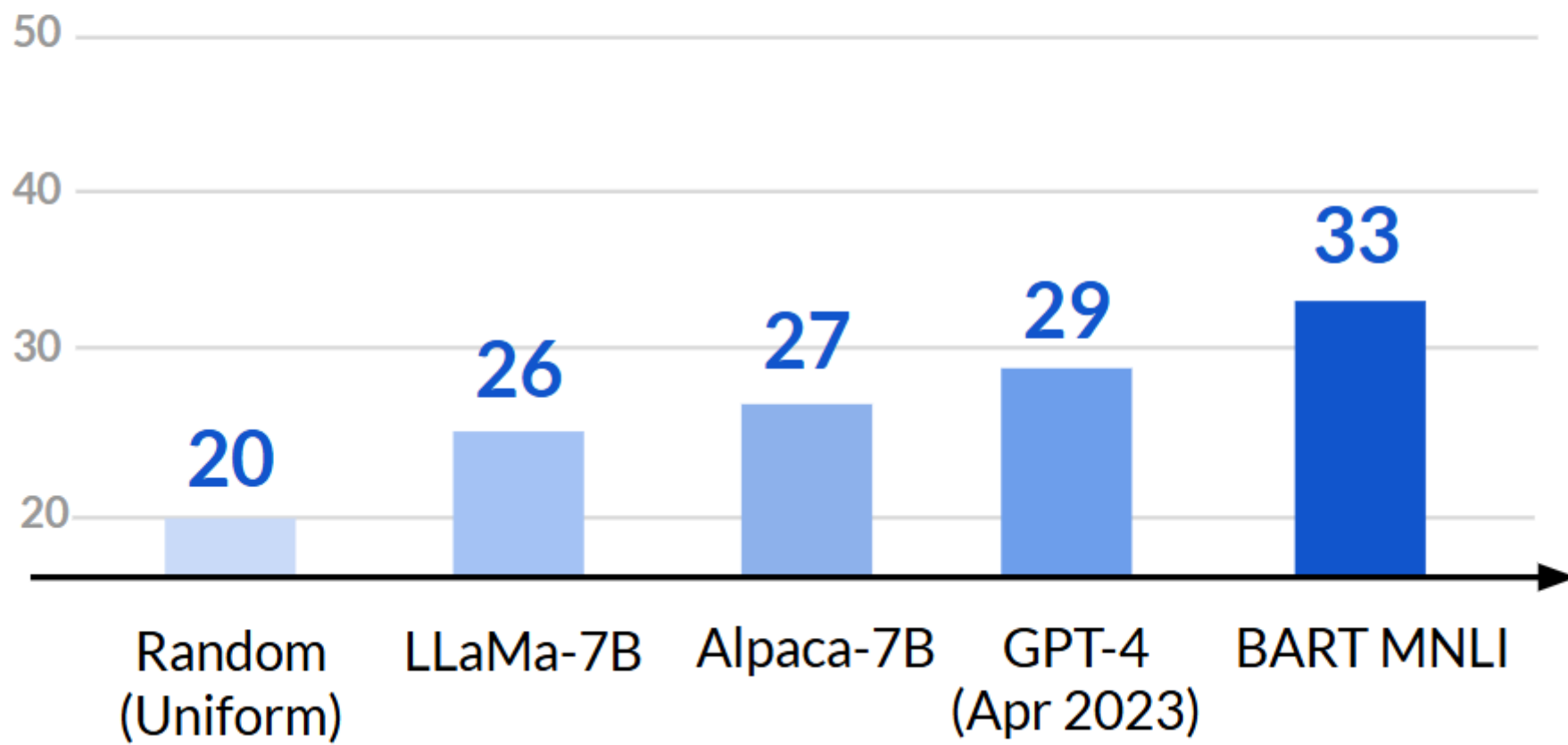
Touching a hot stove $\xrightarrow{\text{Causes}}$ Getting burned

Skill being tested in previous work: *Empirical* knowledge instead of *pure* causal inference.

*Assumption that we explicitly mention in the samples: We suppose a close system of the given variables and correlations.

Model Performance

F1 Score



Scaling Up: LLM-Only Causal Discovery

EFFICIENT CAUSAL GRAPH DISCOVERY USING LARGE LANGUAGE MODELS

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Potential Issues: More Edges than Ground Truth

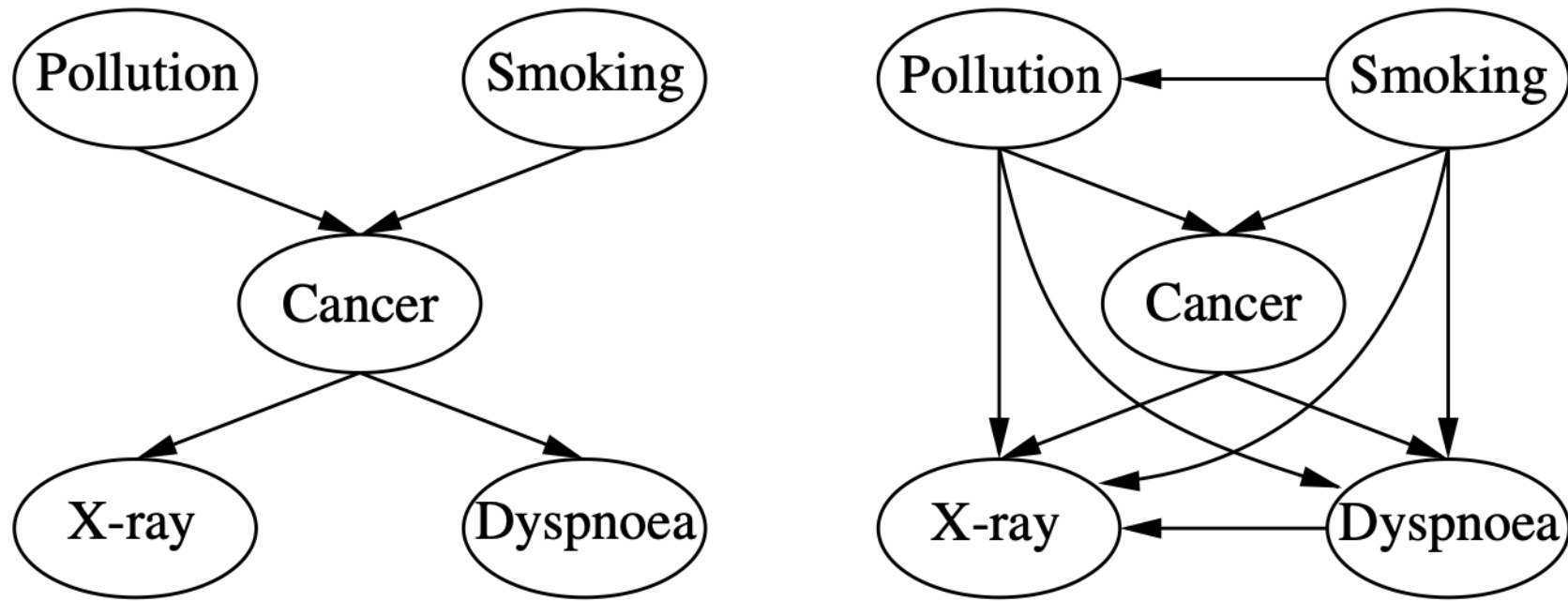


Figure 3: **Left:** Causal graph of Cancer dataset. **Right:** GPT-3.5's estimated causal graph of Cancer

Scaling Up: LLM+Statistical Algorithm

CAUSAL INFERENCE USING LLM-GUIDED DISCOVERY

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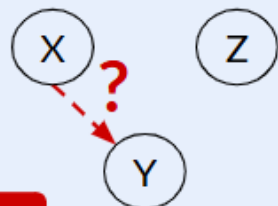
Scaling Up: LLM + Statistical Algorithm

		PC	SCORE	ICA LiNGAM	Direct LiNGAM	NOTEARS	CaMML	Ours (PC+LLM)	Ours (CaMML+LLM)
$N = 250$	Earthquake	0.30±0.44	4.00±0.00	3.20±0.39	3.00±0.00	1.80±0.74	2.00±0.00	0.00±0.00	0.00±0.00
	Cancer	0.00±0.00	3.00±0.00	4.00±0.00	3.60±0.48	2.00±0.00	2.00±0.00	0.00±0.00	0.00±0.00
	Survey	0.50±0.00	3.00±0.00	6.00±0.00	6.00±0.00	3.20±0.39	3.33±0.94	0.00±0.00	3.33±0.94
	Asia	2.33±0.59	5.00±0.00	6.20±0.74	7.00±0.00	4.00±0.00	1.85±0.58	0.00±0.00	0.97±0.62
	Asia-M	2.00±0.00	5.00±0.00	7.60±0.48	6.20±1.16	3.40±0.48	1.00±0.00	0.00±0.00	1.71±0.45
	Child	8.16±1.58	8.80±2.70	12.8±0.97	13.0±0.63	15.0±1.09	3.00±0.00	4.00±0.00	3.53±0.45
	Neuropathic	3.25±0.00	6.00±0.00	13.0±6.16	10.0±0.00	9.00±0.00	10.4±1.95	1.00±0.00	5.00±0.00

Two Main Causal Inference Tasks

Task 1. Causal Discovery

What variable affects what?



Example

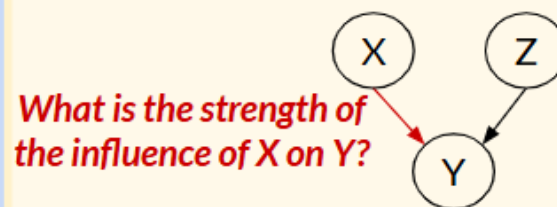
Lots of tech companies' CEOs are college dropouts, does it mean **"dropout" causes "job success"**?

Corr2Cause Paper

(Jin et al, 2024 ICLR)

Task 2. Causal Effect Reasoning

By how much?



Example

How much does COVID vaccination **increase** the survival rate?

Cladder Paper

(Jin* et al, 2023 NeurIPS)



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W



CLadder: Assessing Causal Reasoning in Language Models



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Luigi Gresele*



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Kevin Blin



Fernando Gonzalez



Max Kleiman-Weiner

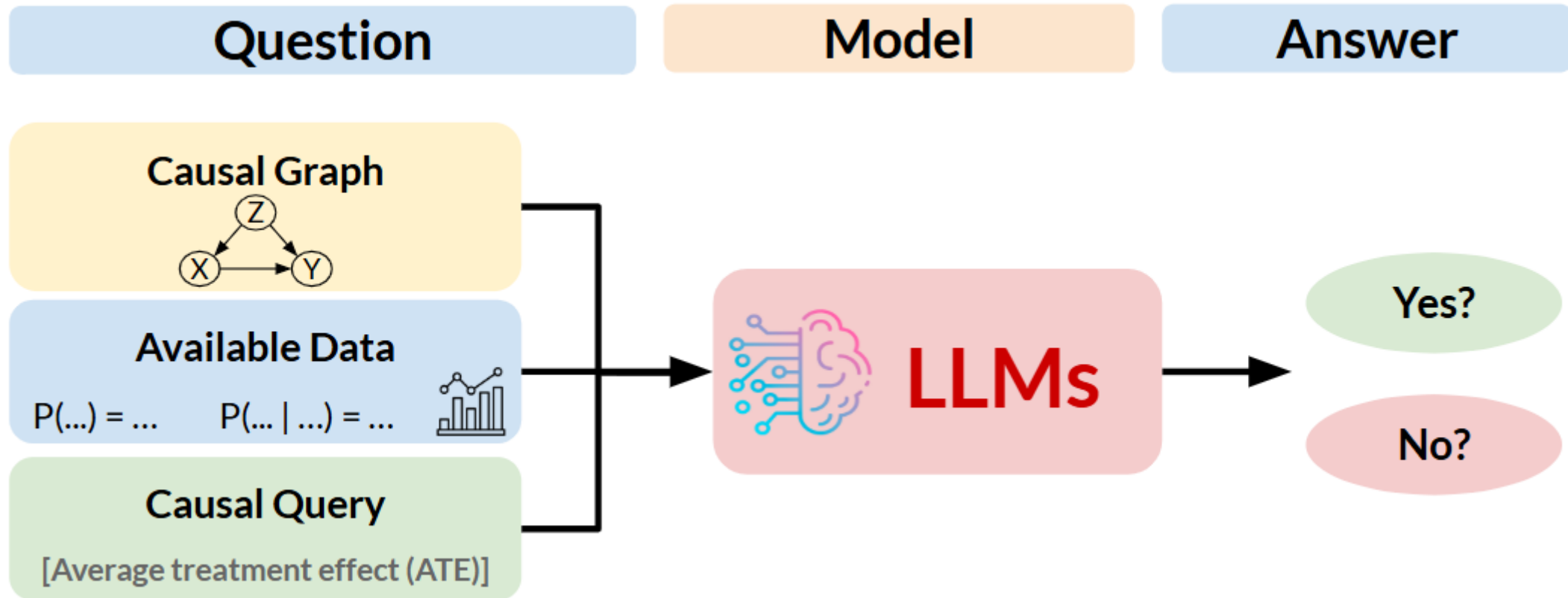


Mrinmaya Sachan

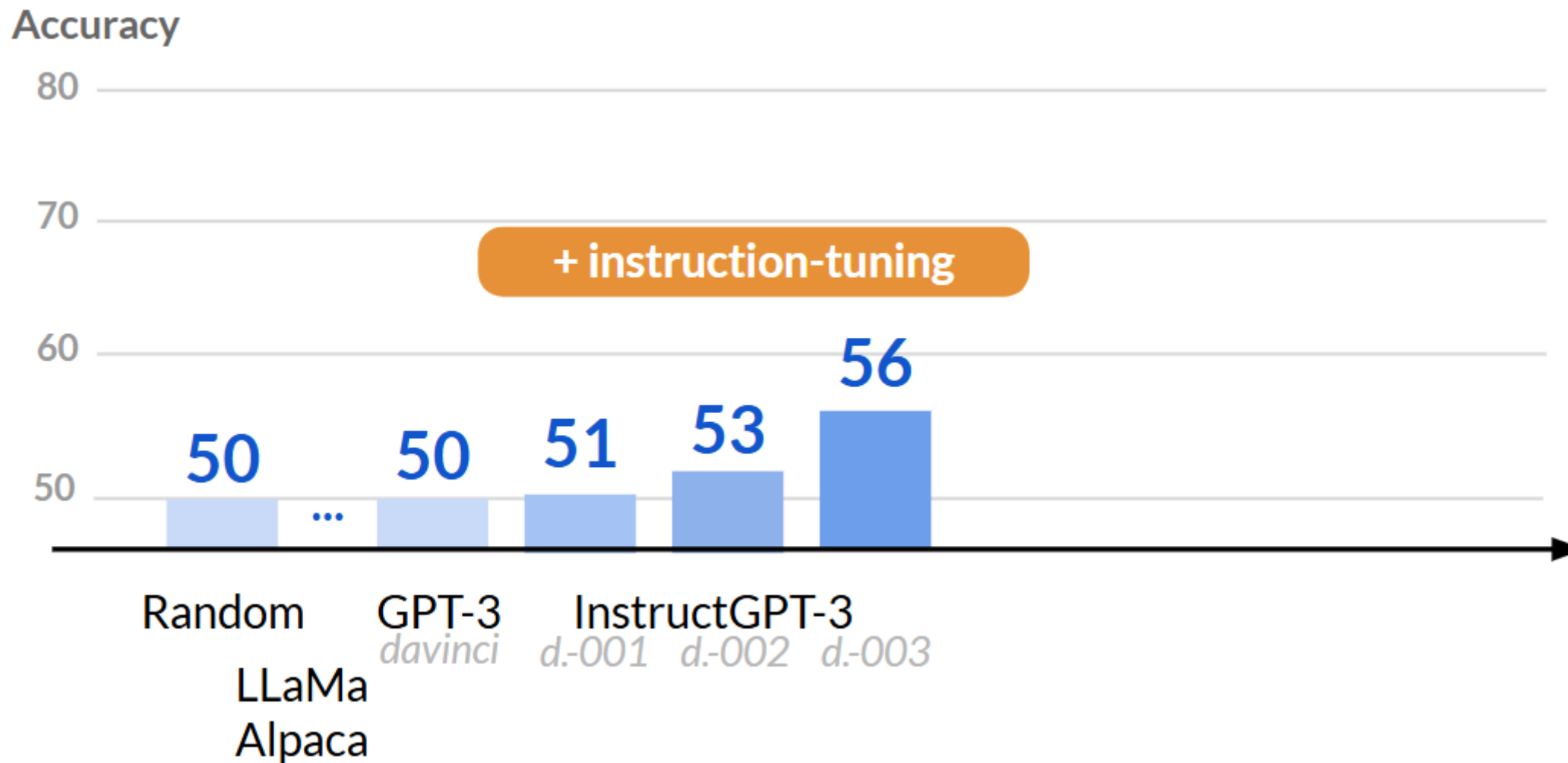


Bernhard Schölkopf

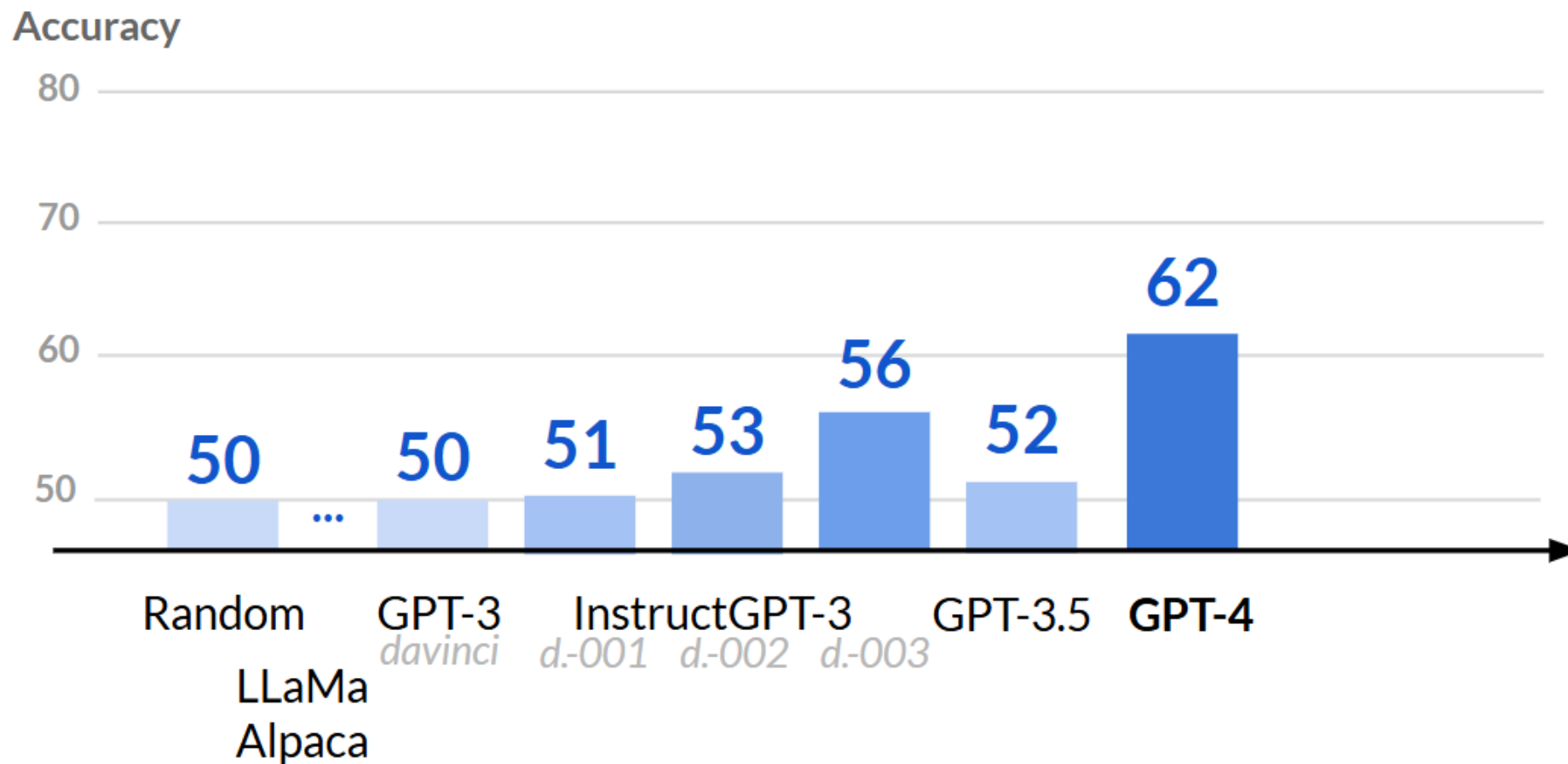
Testing Causal Reasoning in LLMs



Model Performance



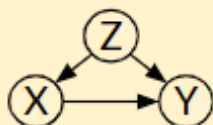
Model Performance



Causal CoT: Causal Chain-of-Thought Prompting

[...] Does vaccination lead to an increase in the survival rate?

Causal Graph
Parsing



Query Formalization

$$\text{ATE } (X \rightarrow Y) \\ = E[Y \mid \text{do}(X=1)] - E[Y \mid \text{do}(X=0)]$$

Statistics Collection

$$P(Z=1)=0.5 \\ P(Y=1 \mid X=1)=...$$

Formal Causal
Inference Steps

E.g., Do-calculus,
Counterfactual prediction

Arithmetics

Final Answer

Yes

Accuracy

80

70

60

50

50

...

50

51

53

56

52

62

8%

70

Random

GPT-3

davinci

InstructGPT-3

d-001

d-002

d-003

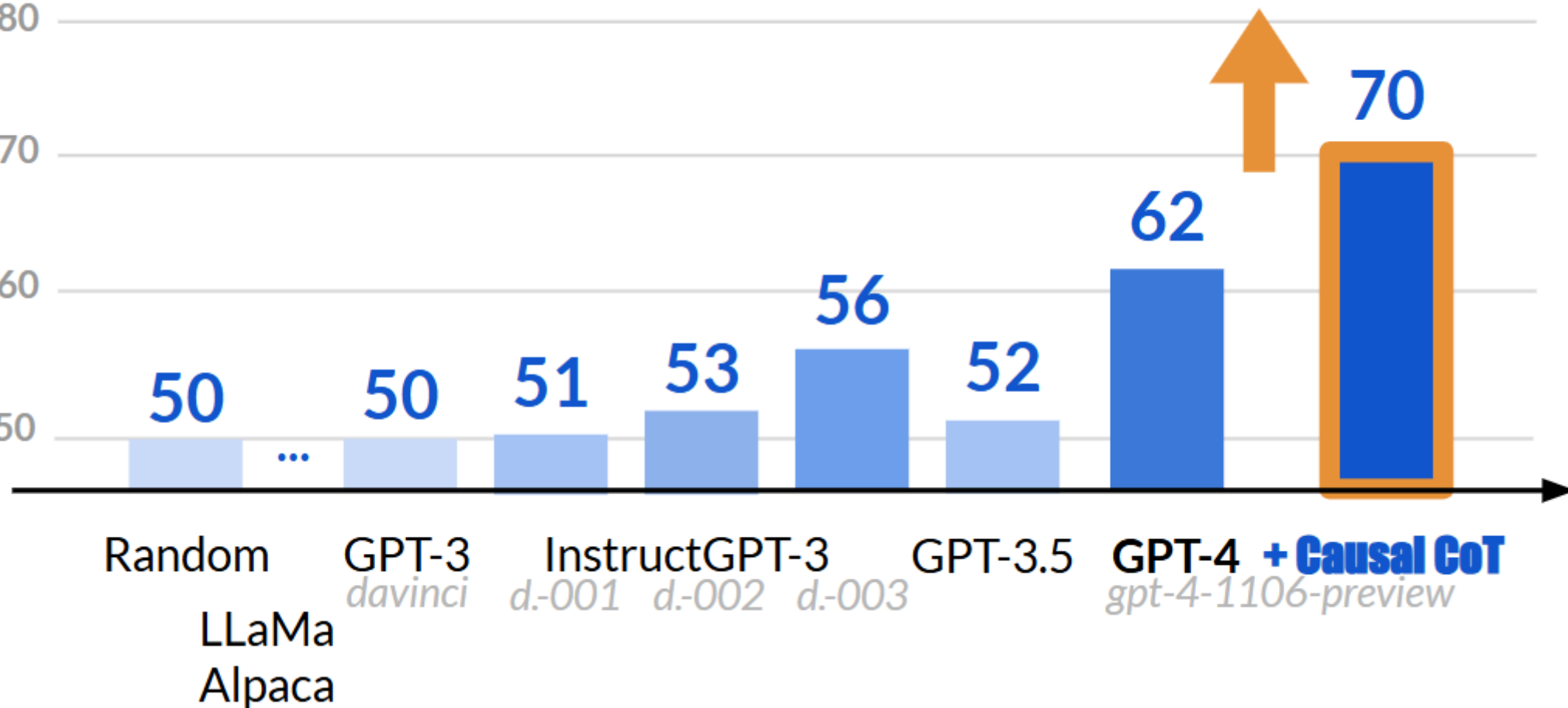
GPT-3.5

GPT-4 + Causal CoT

gpt-4-1106-preview

LLaMa

Alpaca



[...] Does vaccination lead to an increase in the survival rate?

Causal Graph
Parsing



Query Formalization

$$\text{ATE } (X \rightarrow Y) \\ = E[Y \mid \text{do}(X=1)] - E[Y \mid \text{do}(X=0)]$$

Statistics Collection

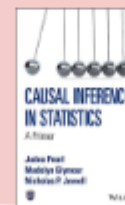
$$P(Z=1)=0.5 \\ P(Y=1 \mid X=1) =$$

**Formal Causal
Inference Steps**

E.g., Do-calculus,
Counterfactual prediction

! Challenging Steps !

⇒ We should really improve LLMs
on formal causal inference!



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Analyzing Human Questioning Behavior and **Causal Curiosity** through Natural Queries

arxiv.org/abs/2405.20318



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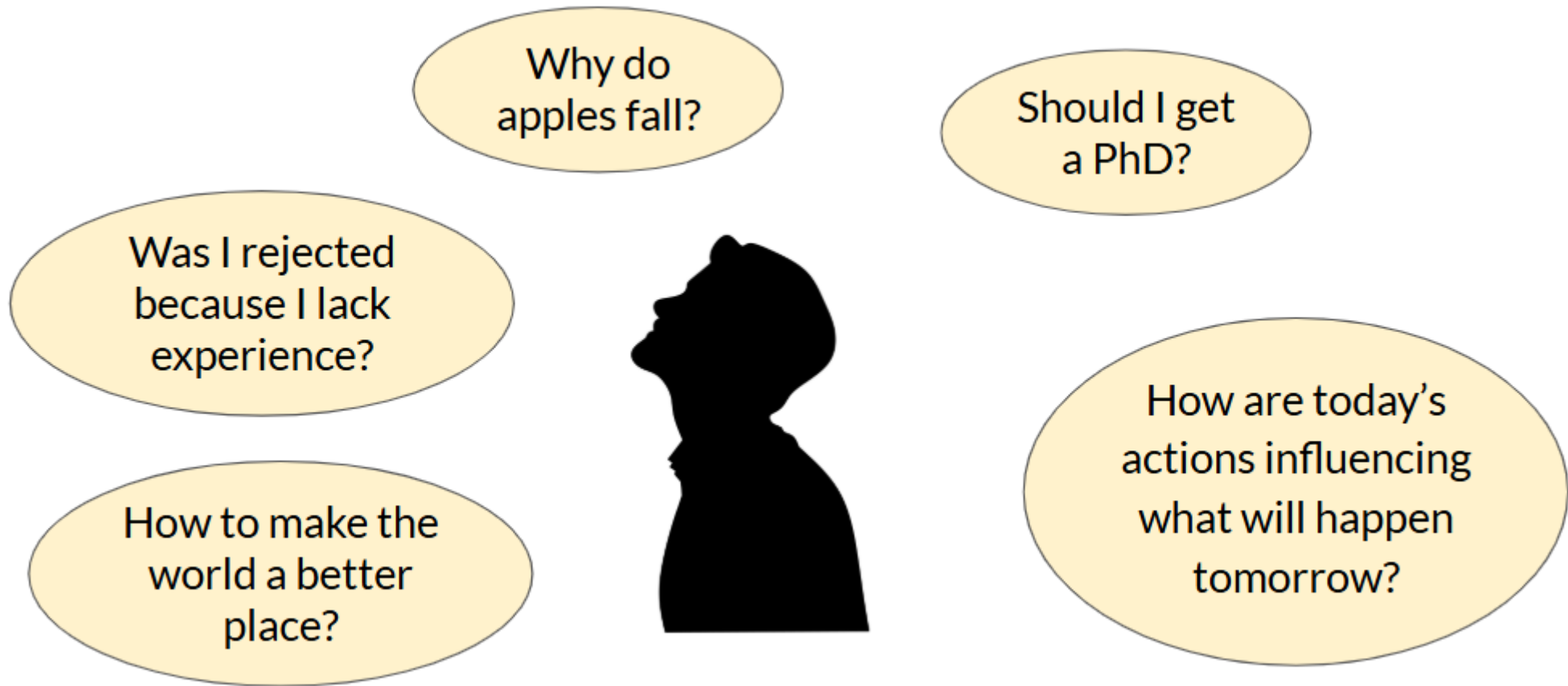


Mrinmaya
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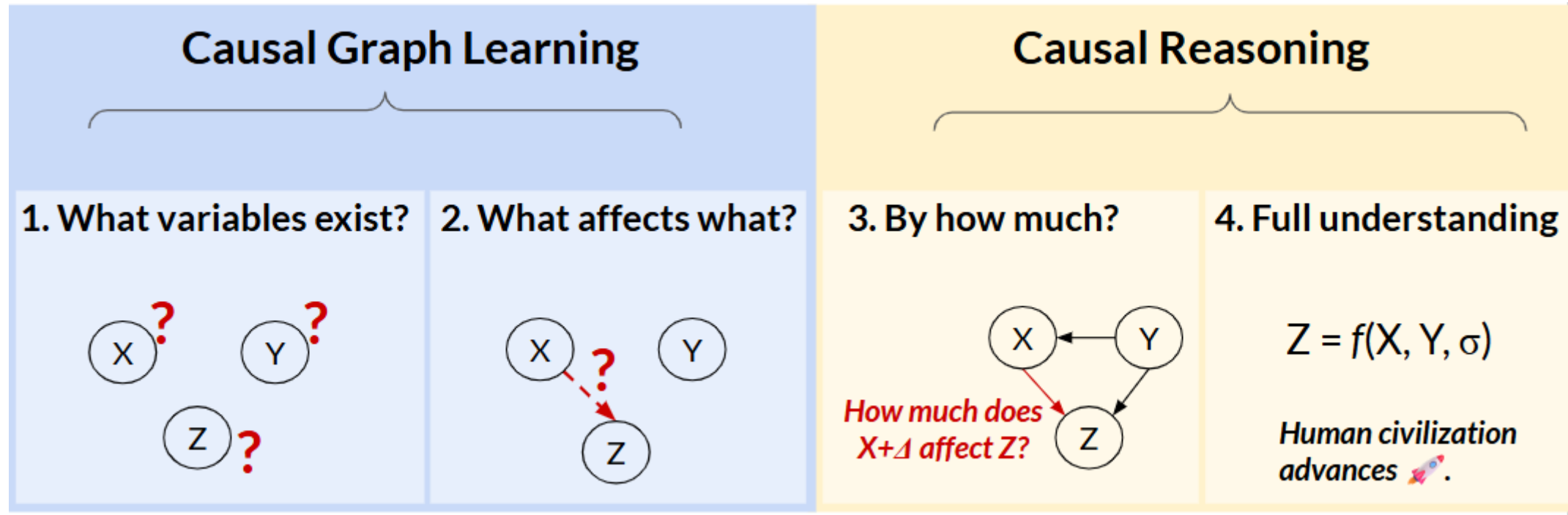


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Jin

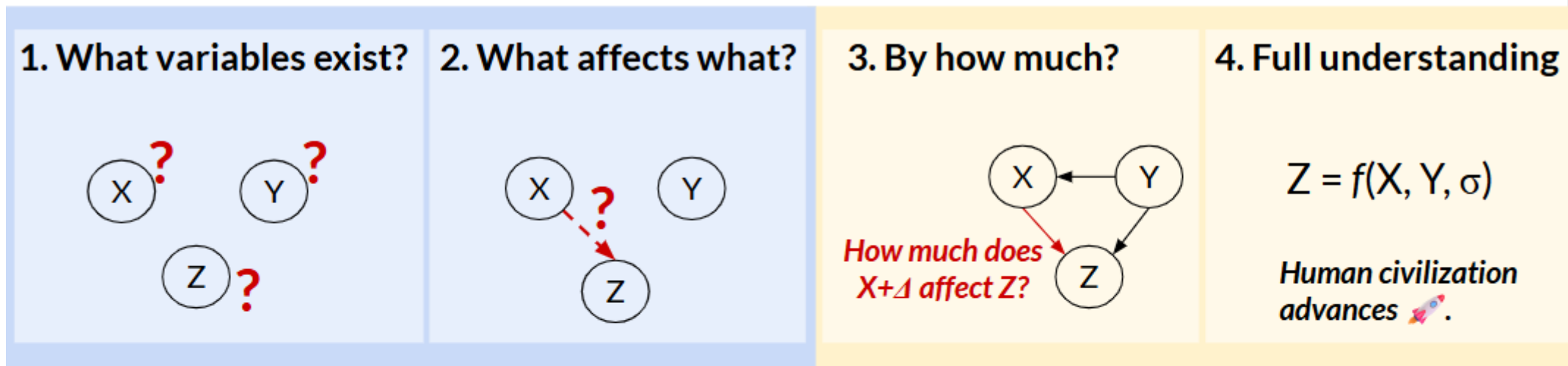
Causal Questions



Taxonomy of Causal Questions



Taxonomy of Causal Questions



1. Asking about the **variables**
2. Asking about the **edges**
3. Asking about the **causal effect**
4. Asking about the **mechanism**

Taxonomy of Causal Questions

- 1 Asking about the **variables**
- 2 Asking about the **edges**
- 3 Asking about the **causal effect**
- 4 Asking about the **mechanism**

What is the change in my income after a PhD?

3

Does smoking interfere with the drug effect?

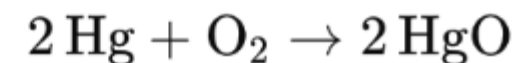
2

What are the key elements needed to start a fire?

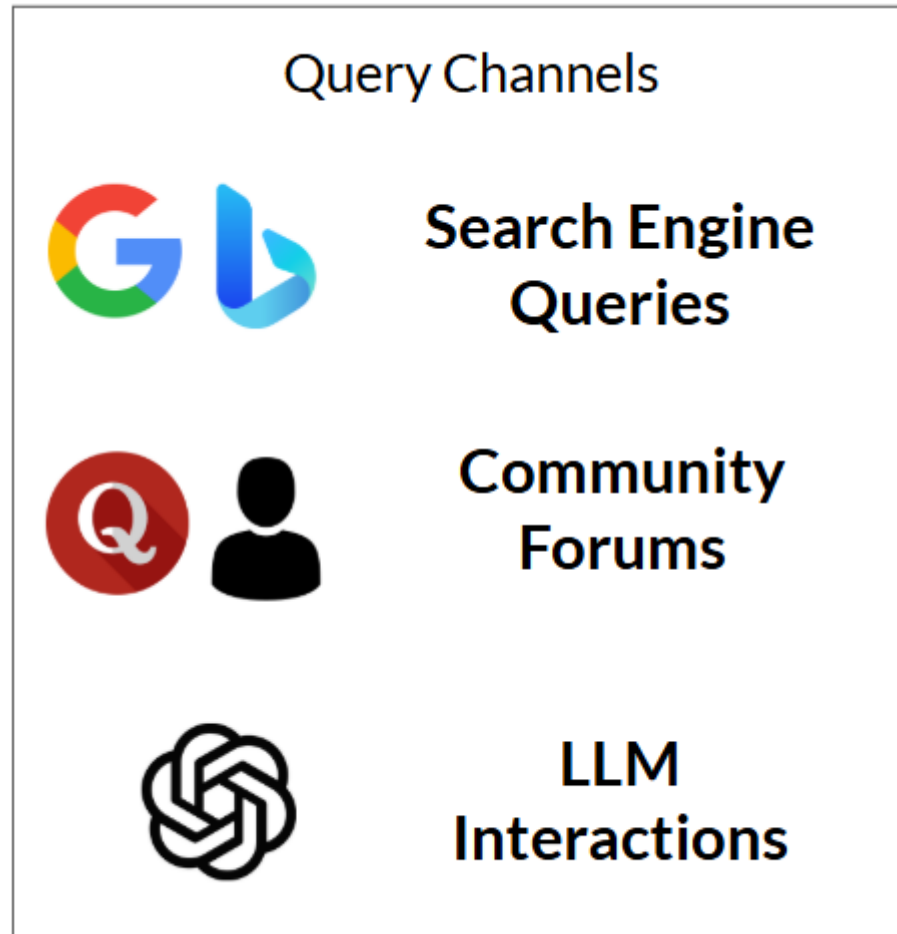
1

How much oxygen (O₂) is needed to oxidize 1 kg of mercury (Hg)?

4



Capturing Human Natural Curiosity



Emergence of
Curiosity



NatQuest data:
13.5K Questions

42%

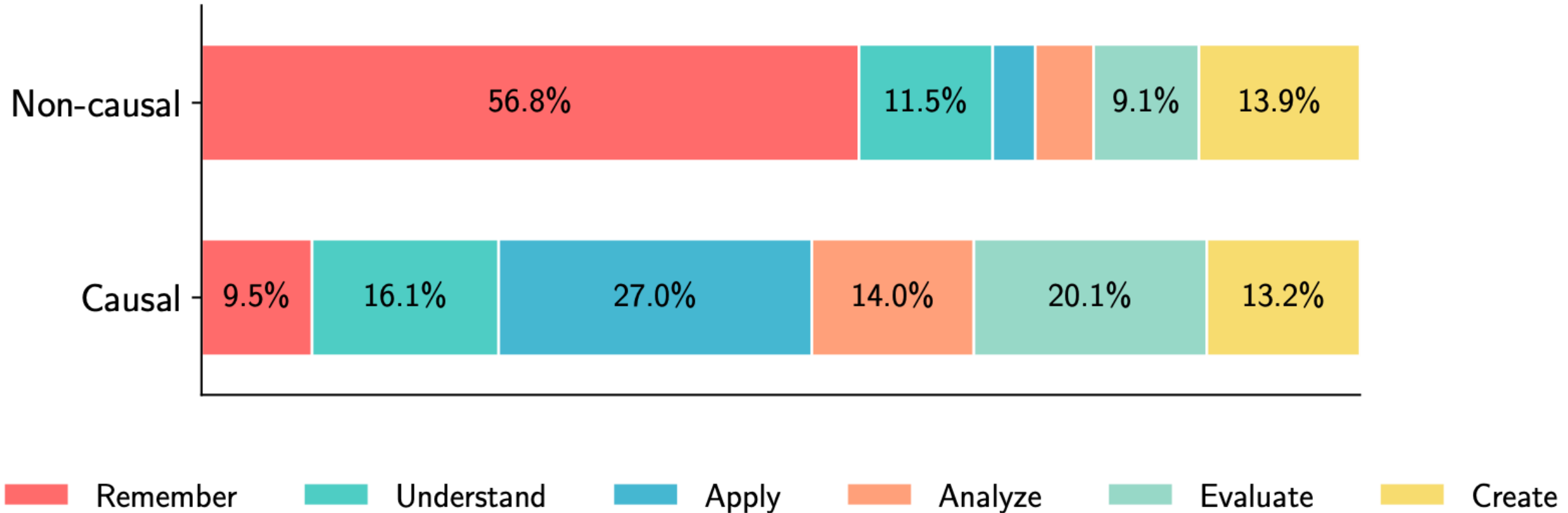
of the natural
questions are causal

Linguistic Properties of Causal Questions

“How” and “Why” questions are often causal

Question Words	Causal	Non-Causal	Overall
How	2,077	636	2,713
Why	635	12	647
What	1,383	2,324	3,707
Who	144	723	867
Where	124	465	589

Levels of Cognitive Skills Needed for Causal Questions



Performance of GPT-4o

GPT-4o score
(by human raters)

Effectiveness

3.83 (out of 5)

Efficiency

2.88 (out of 5)

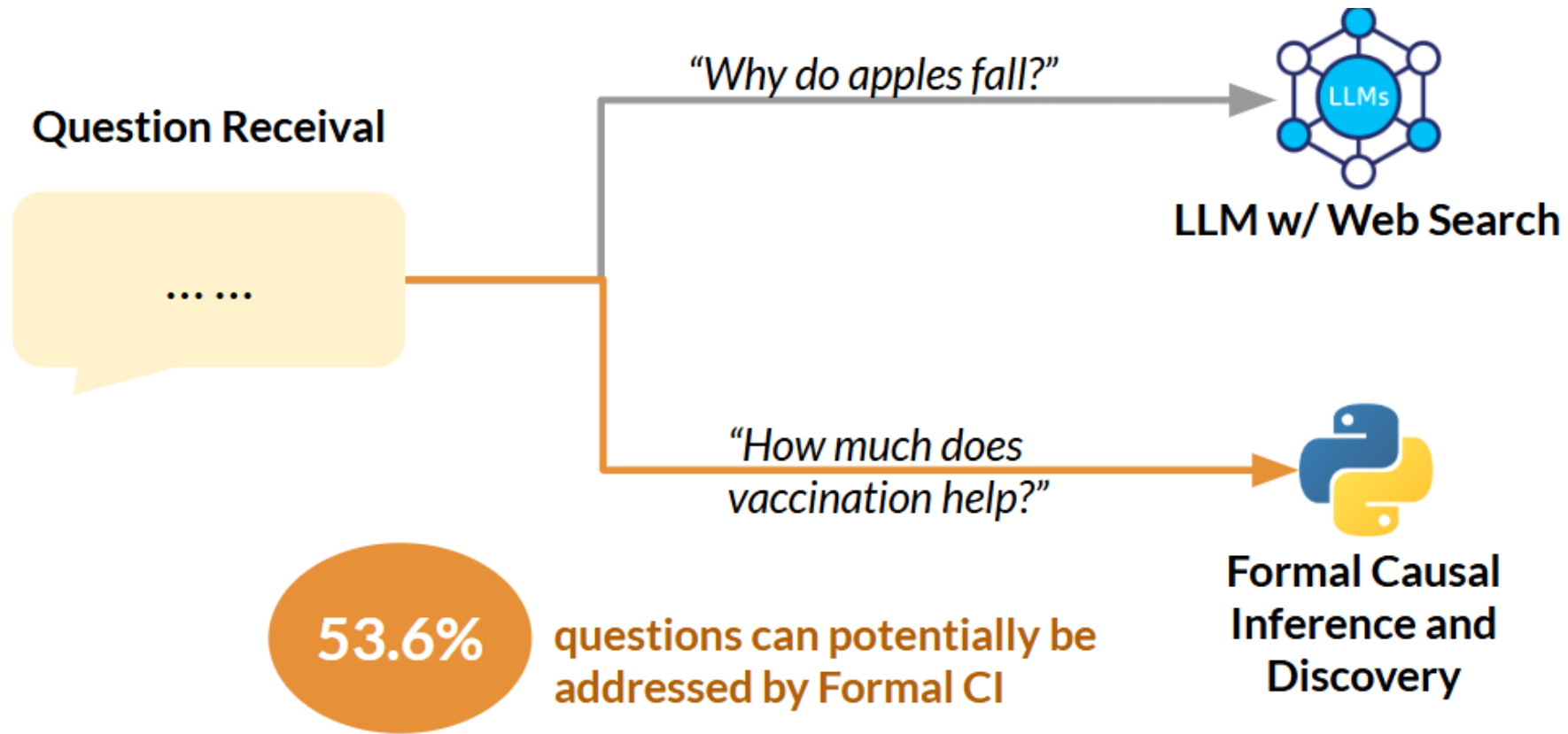
Satisfaction

3.85 (out of 5)

Observations:

- Struggles when asked to foresee the causal effects in the future
- Does well in answering causal questions that require knowledge lookup
- Answers are often overly verbose (low efficiency)

Towards Solving the Causal Questions



Open-ended Questions

- ToMi dataset

ToMi dataset

Examples of stories from the ToMi dataset

1 Oliver dislikes the kitchen
2 Carter entered the porch.
3 Abigail entered the porch.
4 The potato is in the green_suitcase.
5 Abigail exited the porch.
6 Abigail entered the hall.
7 Carter moved the potato to the green_envelope.
8 Oliver entered the hall.

1 Mila entered the closet.
2 Isla entered the closet.
3 Ava entered the closet.
4 The orange is in the blue_container.
5 Isla exited the closet.
6 Isla entered the garage.
7 Ava moved the orange to the green_bathtub.

Round 1

Prompt Template

If you were a user on Reddit exploring the 'Change My View' subreddit, a platform dedicated to facilitating reasoned responses aimed at changing the questioner's perspective, imagine encountering a post titled "{title}" with the following content: "{Body}". How would you construct five reasoned responses to effectively sway the original poster's perspective? Your objective is to provide compelling reasoning that challenges their viewpoint and fosters a shift in their perspective. Approach this task step by step.

Kappa score

- acceptable within the range of 0.61 to 0.80

Table 1: Agreement rate among annotators for “Reasoning Correctness” in the first and second rounds.

Scale	Cohen’s kappa score	Overlap rate
Round 1		
5-point Likert scale	0.3527	45.10%
3-point Likert scale	0.4751	61.73%
Dichotomous scale	0.6618	83.66%
Round 2		
Dichotomous scale	0.7204	86.24%

Round 2

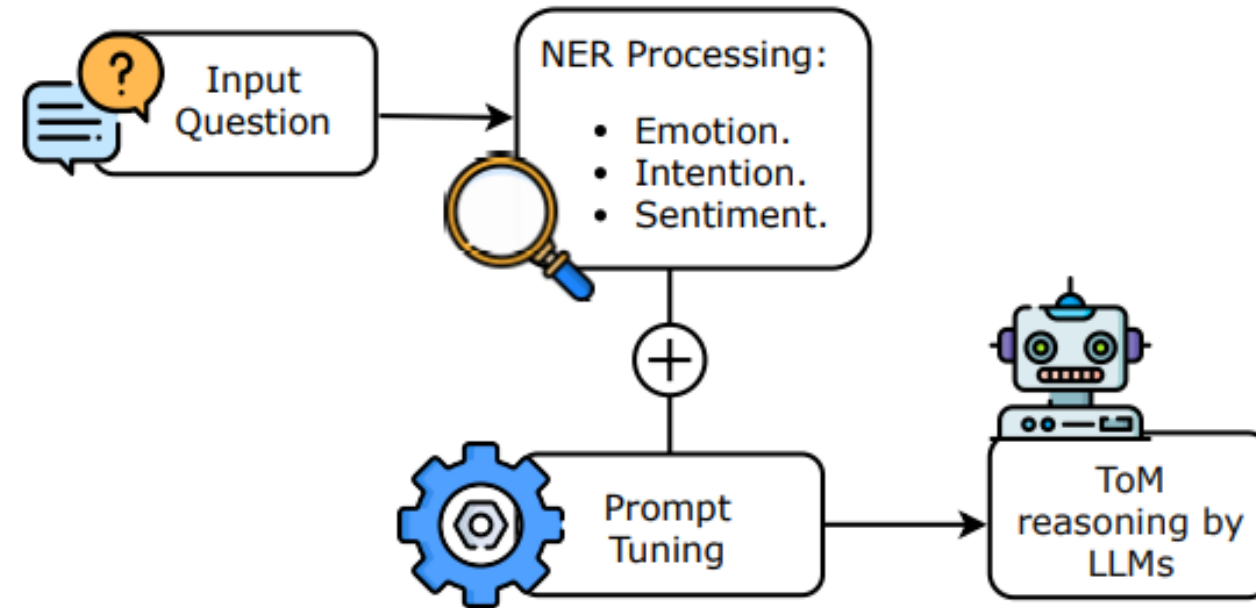


Figure 3: ToM reasoning process via prompt tuning.

Round 2

Prompt Template

If you were a user on Reddit exploring the 'Change My View' subreddit, a platform dedicated to facilitating reasoned responses aimed at changing the questioner's perspective, imagine encountering a post titled "{title}" with the following content: "{Body}". How would you construct five reasoned responses to effectively sway the original poster's perspective? Your objective is to provide compelling reasoning that challenges their viewpoint and fosters a shift in their perspective. Approach this task step by step.

Consider the user's intent "{user's intention}" and emotion. "{user's emotion}", along with the sentiment of the post "{post sentiment}", as you construct your reasoning. Think step by step.

Figure 4: Final prompt template.